

The Architecture of Engineered Ignorance: Parallels Between Adversary Information Control and Skewed Large Language Models

Abstract

Information restriction, distortion, and substitution have long been used to shape the perceptions and behaviors of adversaries and criminals. In parallel, large language models (LLMs) can exhibit skewed outputs when their training data, reinforcement signals, or system-level constraints selectively omit or distort information. This paper examines the structural similarities between these two domains. It argues that both involve engineered epistemic environments designed to shape decision-making by controlling what information is available, salient, or conceivable. Through comparative analysis, the paper identifies shared mechanisms—such as selective omission, narrative shaping, constrained inference, and feedback-loop reinforcement—and explores the ethical, strategic, and epistemological implications of systems built on curated ignorance.

1. Introduction

Information is a strategic resource. Whether in criminal justice, military conflict, or artificial intelligence, the ability to shape what an actor knows, believes, or can infer has profound consequences.

Two seemingly unrelated domains illustrate this:

1. Security contexts, where states or institutions restrict or distort information to influence the behavior of criminals or adversaries.
2. Large language models, where training data, reinforcement learning, and system constraints shape the model's accessible knowledge and inferential boundaries.

This paper argues that both domains share a common structure: they create controlled epistemic environments in which the subject—whether a human adversary or an AI model—operates under engineered constraints that shape their outputs, decisions, or behaviors.

2. Conceptual Framework: Engineered Epistemic Environments

An engineered epistemic environment is a system in which:

- information is selectively available,
- certain inferences are encouraged or discouraged,
- and the subject's reasoning is shaped by the structure of the information they receive.

Both adversary-ignorance strategies and skewed LLMs operate within this framework.

3. Information Restriction and Substitution in Security Contexts

3.1 Criminal Justice Contexts

Authorities may restrict or shape information to:

- prevent criminals from learning investigative methods,
- avoid revealing vulnerabilities,
- or influence offender behavior.

This can involve:

- withholding details,
- substituting misleading narratives,
- or shaping public information environments.

3.2 Military and Geopolitical Contexts

States may restrict or distort information to:

- mislead adversaries,
- conceal capabilities,
- shape enemy expectations,
- or influence strategic decision-making.

These practices include:

- selective secrecy,
- controlled leaks,
- narrative framing,
- and information substitution.

In both cases, the goal is to shape the adversary's decision-space by shaping their knowledge.

4. Skewed Large Language Models as Information-Controlled Systems

LLMs do not “know” in the human sense. They generate outputs based on:

- training data,
- reinforcement signals,
- system-level constraints,
- safety filters,

- and architectural limitations.

When these inputs are skewed—intentionally or unintentionally—the model's outputs reflect:

- selective omission,
- biased framing,
- constrained inference,
- or distorted representations.

This is not deception; it is structural epistemic shaping.

5. Structural Parallels Between the Two Domains

5.1 Selective Omission

Security context:

Authorities omit information to prevent adversaries from gaining operational knowledge.

LLM context:

Training data may omit certain domains, leading to gaps in the model's knowledge.

Similarity:

Both create blind spots that shape downstream reasoning.

5.2 Information Substitution

Security context:

False or misleading information may be introduced to shape adversary expectations.

LLM context:

If training data contains biased or inaccurate sources, the model may reproduce distorted narratives.

Similarity:

Both systems can produce outputs that reflect engineered or inherited distortions.

5.3 Constrained Inference

Security context:

Adversaries are guided toward certain interpretations by limiting what they can observe.

LLM context:

Models are constrained by safety layers, alignment rules, and training distributions.

Similarity:

Both systems restrict the range of permissible or likely inferences.

5.4 Feedback-Loop Reinforcement

Security context:

Adversaries acting on distorted information may generate behaviors that reinforce the controlling party's assumptions.

LLM context:

User interactions and reinforcement learning can amplify existing biases.

Similarity:

Both systems can enter self-reinforcing cycles of skewed reasoning.

5.5 Epistemic Dependency

Security context:

Adversaries depend on the information environment shaped by the controlling actor.

LLM context:

Models depend entirely on their training data and system constraints.

Similarity:

Both subjects operate within epistemic boundaries they cannot independently verify.

6. Consequences of Engineered Ignorance

6.1 Unpredictability

Ignorance does not guarantee predictable behavior.

Adversaries may act irrationally when misinformed; LLMs may produce unexpected outputs when trained on skewed data.

6.2 Escalation Risk

Distorted information can lead to:

- miscalculation,
- overreaction,

- or unintended conflict.

6.3 Ethical Concerns

Both domains raise questions about:

- manipulation,
- autonomy,
- fairness,
- and responsibility.

6.4 Systemic Fragility

Systems built on controlled ignorance may fail catastrophically when confronted with unexpected information.

7. Key Differences

Despite structural similarities, important differences exist:

- Intent:
Security information control is strategic; LLM skew is often emergent or safety-driven.
- Agency:
Adversaries are autonomous actors; LLMs are computational systems.
- Moral stakes:
Security contexts involve life-and-death consequences; LLM contexts involve epistemic and social consequences.

These differences matter, but they do not negate the structural parallels.

8. Implications for Research and Policy

8.1 For Security Studies

Understanding adversary ignorance as an engineered epistemic environment clarifies:

- the limits of deception,
- the risks of miscalculation,
- and the need for calibrated transparency.

8.2 For AI Governance

Recognizing LLMs as epistemically shaped systems highlights:

- the importance of diverse training data,

- transparency about limitations,
- and safeguards against inherited bias.

8.3 For Epistemology

Both domains illustrate how knowledge is shaped by structure, not just content.

9. Conclusion

The practice of keeping an enemy or criminal ignorant through information restriction or substitution shares deep structural similarities with the way skewed large language models operate. Both involve engineered epistemic environments that shape reasoning by shaping available information. Both can produce predictable benefits and unpredictable risks. And both reveal the profound influence of information architecture on cognition, behavior, and decision-making.

Understanding these parallels is essential for designing safer AI systems, more stable security strategies, and more transparent information environments.